MICHIGAN MEDICINE
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Guidelines for Treatment of Native Valve Infective Endocarditis in Adults

Infectious Diseases consultation is STRONGLY recommended

Transition to ORAL antibiotics in endocarditis could be considered for adult patients meeting the following criteria AND requires approval from the multidisciplinary endocarditis treatment team:

Clinical status
- Clinically stable – afebrile, improving WBC
- Blood culture clearance for at least 48 – 72 hrs
- No further surgical intervention planned this admission
- Functional GI tract, able to absorb oral antibiotics
- BMI < 40

Type of endocarditis
- Blood culture with *Staphylococcus aureus, Streptococcus spp, Enterococcus faecalis,* or Coagulase-negative staphylococci, or HACEK organisms
- Organism is susceptible to an oral antibiotic listed in the treatment recommendations below, though this does not preclude use of other regimens in individual cases after discussion with ID and ID pharmacy

Logistic considerations
- **If Linezolid is used, has an appropriate linezolid trough drawn (refer to Linezolid TDM guideline)?**
- **Enrollment into Complex Outpatient Antimicrobial Therapy (COpAT) program for monitoring**
- No psychological or social issues that would prevent adherence to oral antibiotic therapy
- Plans/able follow up with ID at Michigan Medicine
- Has at least one working phone # and ideally also a secondary phone #/emergency contact, confirmed with patient
- Have a local laboratory where patient can get weekly monitoring labs
- Insurance approval for antibiotics
- If possible, deliver meds to bedside prior to patient discharge

Exclusion criteria:

- Patients with surgical indications (e.g., paravalvular abscess, metastatic sites of infections) who are not felt to be good surgical candidates

Empiric Therapy[§]	Pathogens		Subsequent Therapy (Renal Dose Adjustments May Be Necessary)	Duration of Therapy	Comments
Vancomycin IV[4] + **Ceftriaxone** 2 g IV q24h NOTE: **Cefepime** 2 g IV q8h[1] should be used instead of ceftriaxone in burn patients and people who inject drugs	**Viridans group streptococci** **OR** ***Streptococcus gallolyticus (bovis)***	Penicillin MIC ≤ 0.12 mg/L	Preferred: **Ceftriaxone** 2 g IV q24h OR **Penicillin G** 3 million units IV q4h[1,2] Alternative for ceftriaxone allergy: **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h Alternative oral regimen: **Amoxicillin** (refer to renal dosing guidelines)	4 weeks	• Daptomycin monotherapy should NOT be prescribed for Viridans group *Streptococci* • In patients with renal insufficiency, dosing adjustments should be made with PharmD • Consider dalbavancin in patients unable to receive recommended oral or IV antibiotic therapy • Linezolid serum trough monitoring is recommended for patients on >2 weeks of therapy • Amoxicillin oral regimen is a reasonable alternative, especially in patients who have baseline renal impairment; discuss dosing with ID pharmacy • Vancomycin may the most pragmatic option in dialysis patients • 4-week duration indicated only if symptoms of infection <3-month duration.
		Penicillin MIC > 0.12- < 0.5 mg/L	Preferred (if susceptible): **Ceftriaxone** 2 g IV q24h Alternative for Severe PCN Allergy: **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h	4 weeks	
		Penicillin MIC ≥ 0.5 mg/L	Preferred (if susceptible): **Ceftriaxone** 2 g IV q24h Alternative for ceftriaxone allergy: **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h	4-6 weeks	

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	Enterococci strains <u>resistant</u> to ampicillin	<u>Preferred</u>: **Daptomycin** 10 mg/kg IV q24h[1] OR **Vancomycin** IV[4] <u>Preferred oral switch regimen:</u> **Linezolid** 600 mg PO q12h	6 weeks	• Follow baseline and weekly CK with daptomycin therapy • If oral therapy is not chosen, Daptomycin is preferred over vancomycin for definitive outpatient therapy. However, Vancomycin may be more pragmatic in dialysis patients • Linezolid serum trough monitoring is recommended for patients on >2 weeks of therapy
	Enterococci strains <u>resistant</u> to vancomycin and ampicillin	<u>Preferred</u>: **Daptomycin** 10 mg/kg IV q24h[1] <u>Preferred oral switch regimen:</u> **Linezolid** 600 mg PO q12h	> 6 weeks	• Follow baseline and weekly CK with daptomycin therapy. • Combination therapy with daptomycin and ampicillin or ceftaroline may be considered in patients with persistent disease • Linezolid serum trough monitoring is recommended for patients on >2 weeks of therapy
	Staphylococci (MSSA)	<u>Preferred</u>: **Cefazolin** 2 g IV q8h[1] OR **Oxacillin** 2 g IV q4h[2] <u>Cefazolin allergy:</u> **Daptomycin** 10 mg/kg IV q24h[1] <u>Preferred oral switch regimen:</u> **Linezolid** 600 mg PO q12h	6 weeks	• Cefazolin and daptomycin should be used with caution in MSSA-associated CNS. Oxacillin is the preferred option • Ertapenem combination with cefazolin or oxacillin can be considered for persistent bacteremia • Follow baseline and weekly CK with daptomycin therapy • Linezolid serum trough monitoring is recommended for patients on >2 weeks of therapy

Empiric Therapy§	Pathogens	Subsequent Therapy (Renal Dose Adjustments May Be Necessary)	Duration of Therapy	Comments
Vancomycin IV[4] + **Ceftriaxone** 2 g IV q24h NOTE: **Cefepime** 2 g IV q8h[1] should be used instead of ceftriaxone in burn patients and people who inject drugs	**Staphylococci (MRSA)**	Preferred: **Daptomycin** 10 mg/kg IV q24h[1] OR **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h	6 weeks	• Follow baseline and weekly CK with daptomycin therapy • Higher doses of daptomycin could be considered by ID consult team • Combination therapy with ceftaroline could be considered in high-burden of infection or persistent bacteremia beyond 3 days • Consider dalbavancin in patients unable to receive recommended oral or IV antibiotic therapy • If oral therapy is not chosen, Daptomycin is preferred over vancomycin for definitive outpatient therapy. However, Vancomycin may be more pragmatic in dialysis patients • Linezolid serum trough monitoring is recommended for patients on >2 weeks of therapy
	HACEK Group	Preferred: **Ceftriaxone** 2 g IV q24h Alternative: **Ampicillin-sulbactam** 3 g IV q6h[1,5] Alternative for Severe PCN Allergy/Preferred oral switch regimen: **Levofloxacin** 750 mg IV/PO Q24h OR **Ciprofloxacin** 750 mg PO BID	4 weeks	
	***Candida* spp.**	Preferred: **Liposomal amphotericin B** 3-5 mg/kg IV q24h + **Flucytosine**[1] 25 mg/kg PO q6h Alternative for Intolerance to Liposomal Amphotericin B or Flucytosine: **Micafungin** 150 mg IV q24h	At least 6 weeks	• Following initial therapy with IV antifungal agent, long-term suppression with an oral azole may be considered for sensitive pathogens. • Flucytosine may cause myelosuppression and therefore a CBC should be routinely obtained. Consider risk versus benefit of use especially in patients with renal insufficiency. • Flucytosine therapeutic drug monitoring is recommended in all patients - peak level should be drawn after 3-5 days. Goal peak 20-80 mg/L. • *Candida parapsilosis* demonstrates innately higher MICs to the echinocandins and thus empiric use of micafungin for this organism is not preferred.
	Culture negative endocarditis	**Daptomycin** 10 mg/kg IV + **Ceftriaxone** 2 g IV q24h	4-6 weeks	• Receipt of antibiotics prior to obtaining cultures is the most common cause of culture negative IE. There are many infectious and non-infectious causes. An evaluation of epidemiological factors, history of prior cardiovascular infections, exposure to antimicrobials, clinical course, severity, and extracardiac sites of infection should be performed to help guide diagnosis and treatment. • In patients with renal insufficiency, dosing adjustments should be made with PharmD. • Cefepime 2 g IV q8h[1] should be used instead of ceftriaxone in burn patients and IV drug users for empiric coverage of *Pseudomonas*.

§ Prior to confirmation of pathogen

1. Refer to Antimicrobial Dosing Recommendations for dose adjustments in renal dysfunction
2. If candidate for outpatient therapy, may consider administration via continuous infusion (same daily dose)
3. Please refer to the Aminoglycoside Dosing in Adult Patients for guidance on aminoglycoside dosing.
4. Please refer to the Vancomycin Nomogram for guidance on vancomycin dosing and monitoring.
5. Because of the requirement for frequent dosing and the inability to administer via continuous infusion, these drugs are not recommended for home administration
6. The risk of gentamicin and rifampin toxicity outweighs potential benefits for routine use in endocarditis patients based on current primary literature. The 2015 IDSA endocarditis guidelines recommend rifampin and/or gentamicin for some patients with native valve endocarditis, which pre-date literature evaluating the efficacy and safety of these therapies

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References

	Citation
1	Iversen K, Ihlemann N, Gill SU, et al. Partial Oral versus Intravenous Antibiotic Treatment of Endocarditis. *N Engl J Med*. 2019;380(5):415-424. doi:10.1056/NEJMoa1808312
2	Freling S, Wald-Dickler N, Banerjee J, et al. Real-World Application of Oral Therapy for Infective Endocarditis: A Multicenter, Retrospective, Cohort Study. *Clin Infect Dis*. 2023;77(5):672-679. doi:10.1093/cid/ciad119

Revision History

Date	**Revision**
03/2021	Updated vancomycin dosing & hyperlinks
01/2024	Updated oral step-down options
11/2024	Revisions throughout
7/2025	Updated and revised oral switch options

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source